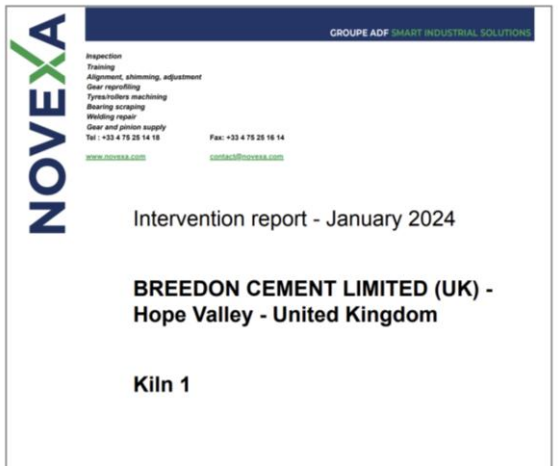
Problem statement


GROUPE ADF SMART INDUSTRIAL SOLUTIONS

NOVEXA

Inspection
Training
Alignment, shimming, adjustment
Gear reprofiling
Tyres/rollers machining
Bearing scraping
Welding repair
Gear and pinion supply

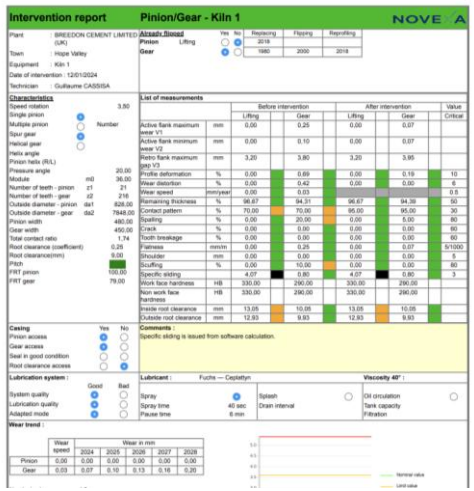
Tel : +33 4 75 25 14 18 Fax: +33 4 75 25 16 14

www.novexa.com contact@novexa.com

Intervention report - January 2024

BREEDON CEMENT LIMITED (UK) - Hope Valley - United Kingdom

Kiln 1


Intervention report — Pinion/Gear - Kiln 1 — NOVEXA

Plant : BREEDON CEMENT LIMITED (UK)
Town : Hope Valley
Equipment : Kiln 1
Date of intervention : 12/01/2024
Technician : Guillaume CASSISA

Already Slipped — Yes / No — Replacing — Flipping — Reprofiling
Pinion — Lifting — 2018
Gear — 1980 — 2000 — 2018

Characteristics
Speed rotation 3,80
Single pinion / Multiple pinion / Spur gear / Helical gear / Helix angle / Pinion helix (R/L) / Number
Pressure angle 20,00
Module m3 36,00
Number of teeth - pinion z1 21
Number of teeth - gear z2 218
Outside diameter - pinion da1 828,00
Outside diameter - gear da2 7848,00
Pinion width 480,00
Gear width 450,00
Total contact ratio 1,74
Root clearance (coefficient) 0,25
Root clearance(mm) 9,00
Pitch
FRT pinion 100,00
FRT gear 79,00

List of measurements

		Before intervention Lifting	Before intervention Gear	After intervention Lifting	After intervention Gear	Value Critical
Active flank maximum wear V1	mm	0,00	0,25	0,00	0,07	
Active flank minimum wear V2	mm	0,00	0,10	0,00	0,07	
Retro flank maximum wear V3	mm	3,20	3,80	3,20	3,95	
Profile deformation	%	0,00	0,89	0,00	0,19	10
Wear distortion	%	0,00	0,42	0,00	0,00	6
Wear speed	mm/year	0,00	0,03			0,5
Remaining thickness	%	96,67	94,31	96,67	94,38	50
Contact pattern	%	70,00	70,00	95,00	95,00	30
Spalling	%	0,00	20,00	0,00	5,00	80
Crack	%	0,00	0,00	0,00	0,00	60
Tooth breakage	%	0,00	0,00	0,00	0,00	60
Fatness	mm/m	0,00	0,25	0,00	0,07	0,1000
Shoulder	mm	0,00	0,00	0,00	0,00	5
Scuffing	%	0,00	10,00	0,00	0,00	80
Specific sliding		4,07	0,80	4,07	0,80	3
Work face hardness	HB	330,00	290,00	330,00	290,00	
Non work face hardness	HB	330,00	290,00	330,00	290,00	
Inside root clearance	mm	13,05	10,05	13,05	10,05	
Outside root clearance	mm	12,93	9,93	12,93	9,93	

Casing — Yes / No
Pinion access
Gear access
Seal in good condition
Root clearance access

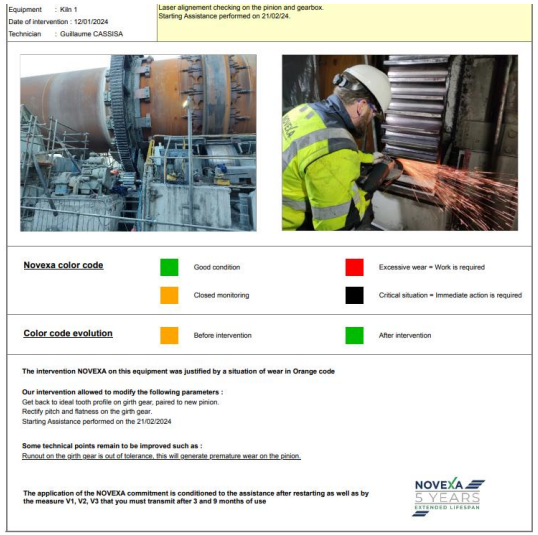
Comments : Specific sliding is issued from software calculation.

Lubrication system : — Good / Bad
System quality
Lubrication quality
Adapted mode

Lubricant : Fuchs — Ceplattyn — Viscosity 40° :
Spray — 40 sec — 6 min — Splash — Drain interval — Oil circulation — Tank capacity — Filtration

Wear trend :

Wear speed	Wear in mm 2024	2025	2026	2027	2028
Pinion	0,00	0,00	0,00	0,00	0,00
Gear	0,03	0,07	0,10	0,13	0,16	0,20


Equipment : Kiln 1
Date of intervention : 12/01/2024
Technician : Guillaume CASSISA

Laser alignement checking on the pinion and gearbox.
Starting Assistance performed on 21/02/24.

Novexa color code

Good condition
Excessive wear = Work is required
Closed monitoring
Critical situation = Immediate action is required

Color code evolution

Before intervention
After intervention

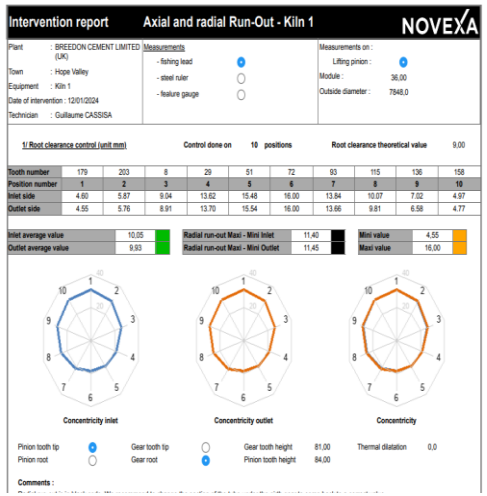
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Some technical points remain to be improved such as :
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NOVEXA 5 YEARS EXTENDED LIFESPAN


Intervention report — Axial and radial Run-Out - Kiln 1 — NOVEXA

Plant : BREEDON CEMENT LIMITED (UK)
Town : Hope Valley
Equipment : Kiln 1
Date of intervention : 12/01/2024
Technician : Guillaume CASSISA

Measurements
- fishing lead
- steel ruler
- feaure gauge

Measurements on :
Lifting pinion :
Module : 36,00
Outside diameter : 7848,0

1/ Root clearance control (unit mm) — Control done on 10 positions — Root clearance theoretical value 9,00

Tooth number	179	203	8	29	51	72	93	115	136	158
Position number	1	2	3	4	5	6	7	8	9	10
Inlet side	4.60	5.87	9.04	13.62	15.48	16.00	13.84	10.07	7.02	4.97
Outlet side	4.55	5.76	8.91	13.70	15.54	16.00	13.66	9.81	6.58	4.77

Inlet average value 10,05
Outlet average value 9,93
Radial run-out Maxi - Mini Inlet 11,40
Radial run-out Maxi - Mini Outlet 11,45
Mini value 4,55
Maxi value 16,00

Concentricity inlet — Concentricity outlet — Concentricity

Pinion tooth tip / Pinion root
Gear tooth tip / Gear root
Gear tooth height 81,00
Pinion tooth height 84,00
Thermal dilatation 0,0

Comments :

Many reports with A LOT of information

Different types of info (text, tables, plots, images)

How to find precise information efficiently?

Large Language Models - Limitations

- Models like ChatGPT, Claude, or others…
- Memory is limited (can not give MANY pdfs)
- Answer can vary because of noise

Solution → Using Retreival-Augmented Generation (RAG)

How RAG Works

Phase 1: Offline preparation

Ingest, process and index your documents

1 PDF Reports



Collect technical reports

2 Extract & Chunk



Extract text and split into chunks

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Phase 2: Question-time inference

Retrieve relevant context and generate an answer

4 User Question



A user asks a question

5 Retrieve Relevant Chunks



Find the most relevant passages

6 LLM Answer + Sources



Generate an answer grounded in the retrieved passages

Answer + page references

"The equipment is rated for 200°C … (pp. 12–14)."

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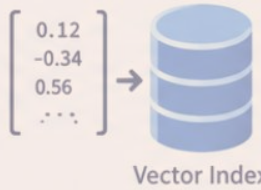
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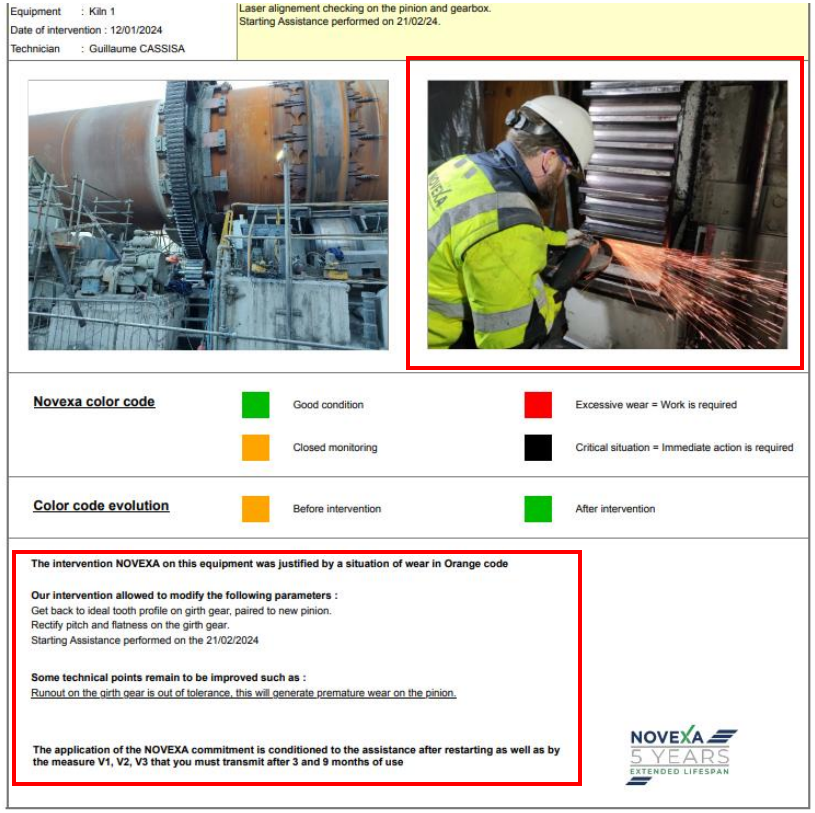
Answer + page references

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2- Extract and Chunk

2.1 - Extraction


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NOVEXA 5 YEARS EXTENDED LIFESPAN

PyMuPDF

Each type is extracted differently (next)



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Provided by PyMuPDF:

Type: Graphics
Position: (x,y)

Type: Text
Position: (x,y)

2- Extract and Chunk

2.1 – Extraction (cont’d)

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For text, PyMuPDF provide everything needed (content + position)

2- Extract and Chunk

2.1 – Extraction (cont'd)

Images on the other hand need to be transformed into useful text



Trained for generating image descriptions

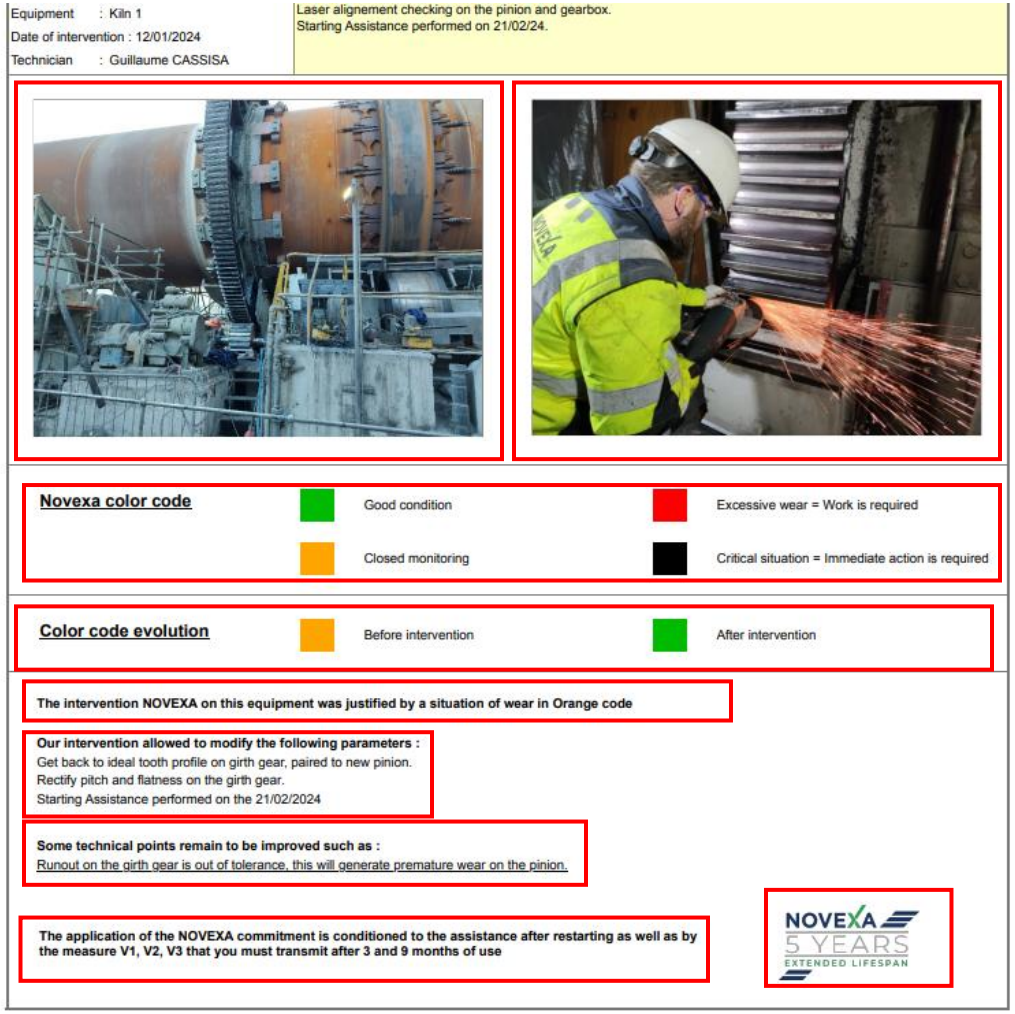
Qwen3-VL

An engineer using a circular chainsaw on a pinion.

NOTE: Since we know the positions of images/text, we also associate images with nearby text to give context to the image.

2- Extract and Chunk

2.2 – Chunking



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NOVEXA 5 YEARS EXTENDED LIFESPAN

Each chunk is linked to what we extracted in 2.1

Chunking is done in a smart way (not just character limits)

But how to make these chunks searchable? (Next)

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Vector Index

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A user asks a question

5 Retrieve Relevant Chunks



Find the most relevant passages

6 LLM Answer + Sources



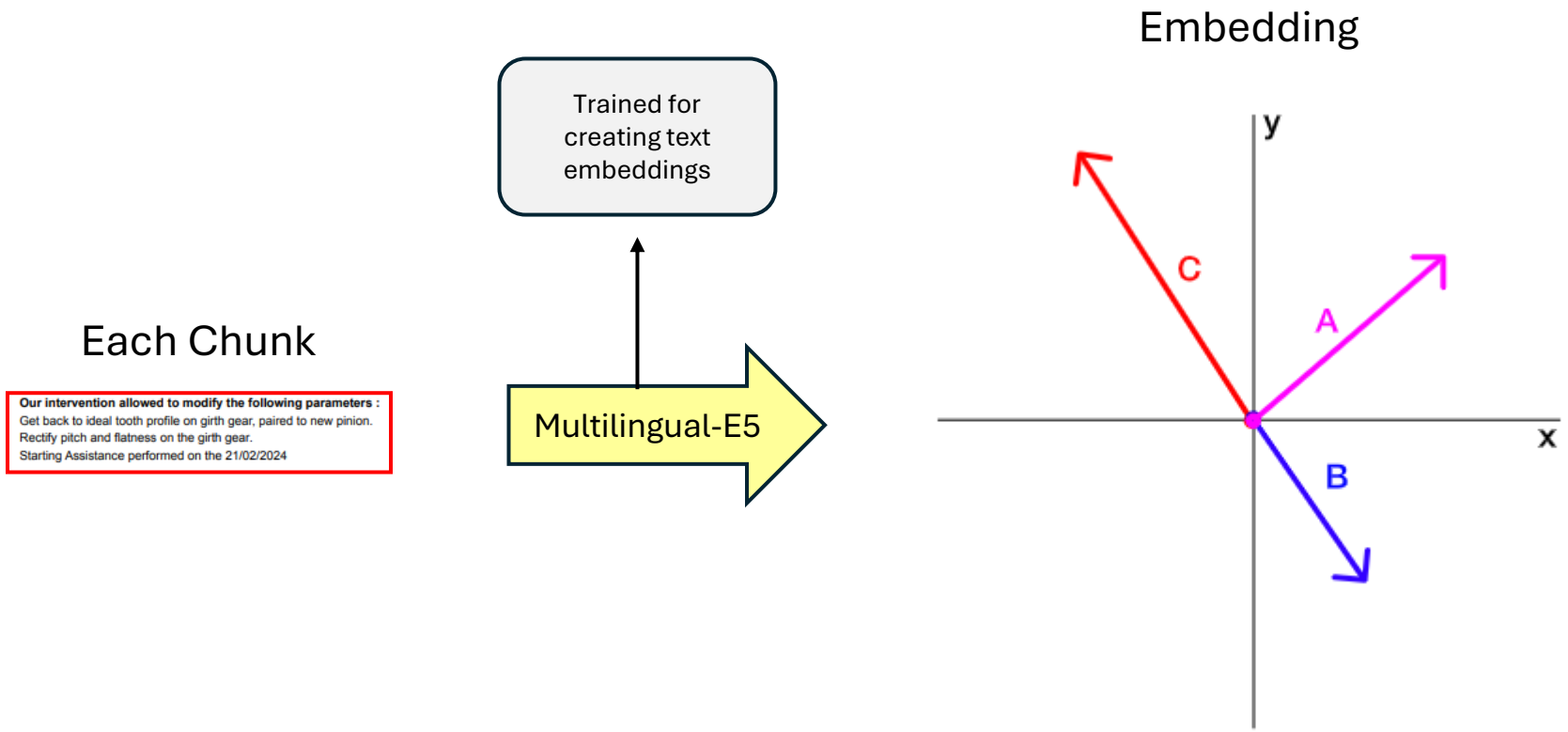
Generate an answer grounded in the retrieved passages

Answer + page references

"The equipment is rated for 200°C … (pp. 12–14)."

RAG = Retrieve the right information first, then let the LLM **answer** using that evidence.

3- Embeddings and Index



NOTE: Using E5 means that the embeddings understand the text!

NOTE: After, we create a searchable index, where for a given index, we can find the corresponding embedding, hence the chunk in one of our PDFs.

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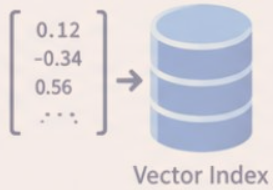
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How does the equipment perform in high temperatures?



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Generate an answer grounded in the retrieved passages

Answer + page references

"The equipment is rated for 200°C … (pp. 12–14)."



RAG = Retrieve the right information first, then let the LLM **answer** using that evidence.

4- User Question

Two possibilities

Text only

Where and when was the worst diameter reported?

Text + Image

Here is a piece I have, what are the recommended actions?



4- User Question

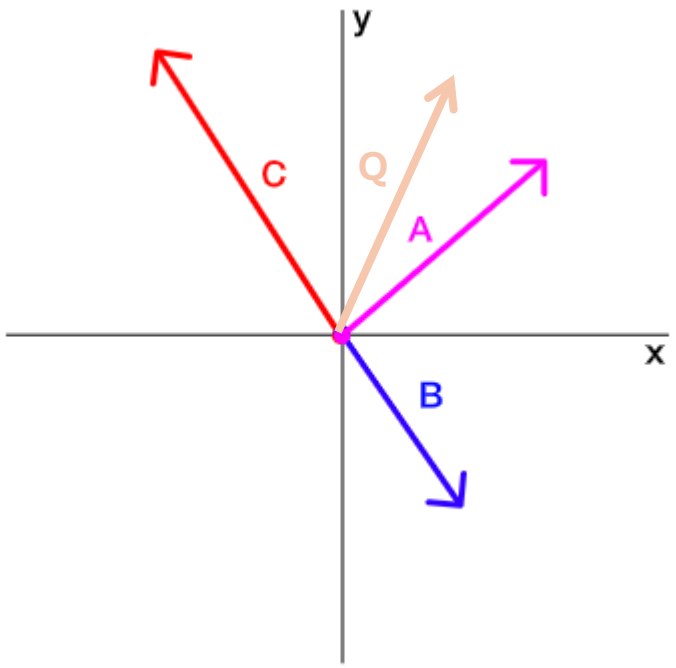
Text only

Where and when was the worst diameter reported?

Same as earlier

Multilingual-E5

Embedding



4- User Question

Text + Image

Here is a piece I have, what are the recommended actions?



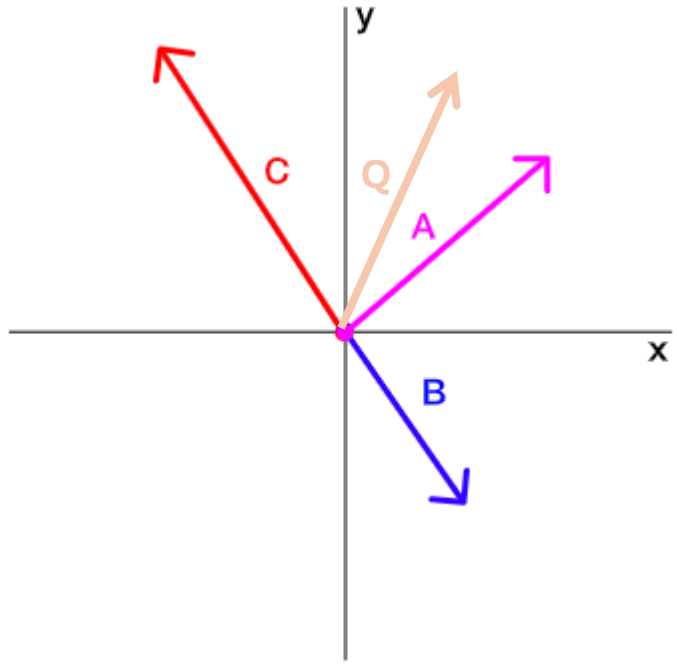
Same as earlier

Qwen3-VL

Image description

Multilingual-E5

Embedding



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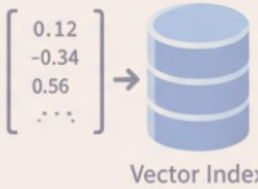
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"The equipment is rated for 200°C … (pp. 12–14)."

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5- Retrieval

Retrieval: is the act of finding what chunks are useful for a given question.

Many techniques exist for retrieval. In this work, we combine the efforts of three techniques.

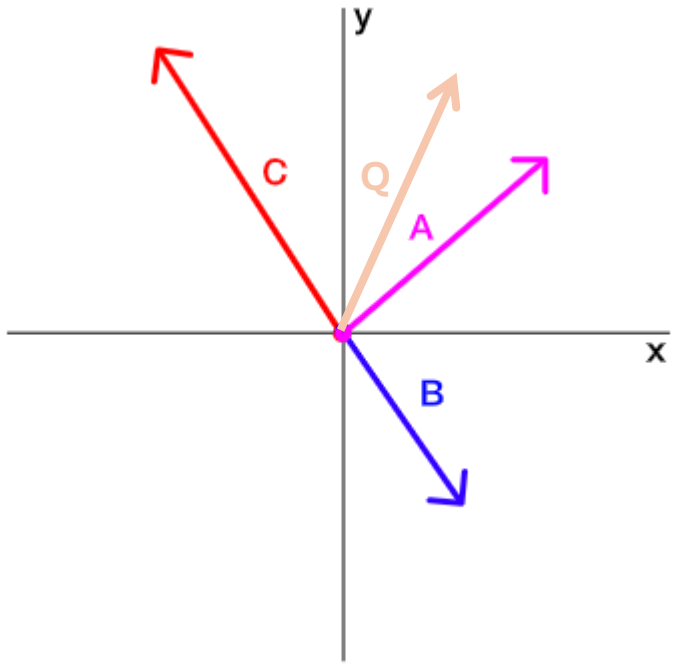
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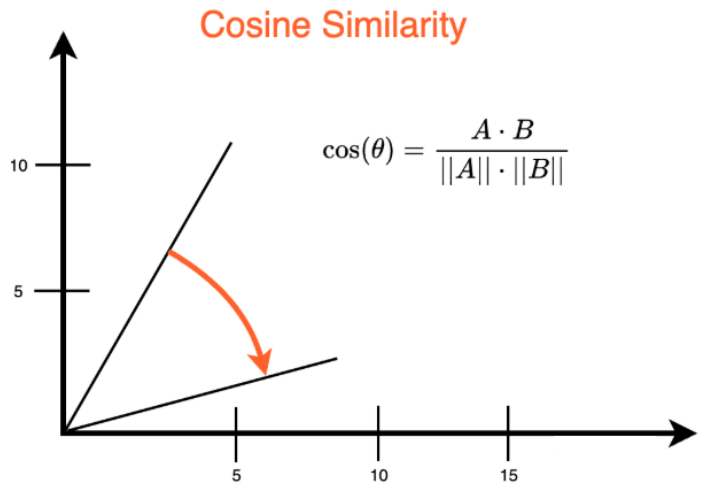
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Many techniques exist for retrieval. In this work, we combine the efforts of three techniques.

E5 semantic retrieval

- Using the embedding vectors previously discussed.
- Find the chunks that have the « closest » vector compared to the question-vector.





Ranking (closest 50)

Limitation: Would miss exact words

5- Retrieval

Retrieval: is the act of finding what chunks are useful for a given question.

Many techniques exist for retrieval. In this work, we combine the efforts of three techniques.

E5 semantic retrieval

- Using the embedding vectors previously discussed.
- Find the chunks that have the « closest » vector compared to the question-vector.

Ranking (closest 50)

BM25 lexical retrieval

- Looking for exact terms in the questions that exist in chunks

Ranking (best 50)

Metadata matching

- Checking if parts of the question appear in the pdf name, section titles, chunk type, etc.

Ranking (best 50)

But how do we combine different rankings?

5- Retrieval

But how do we combine different rankings?

E5 semantic retrieval	BM25 lexical retrieval	Metadata matching
Rank 1: Chunk 15	Rank 1: Chunk 3	Rank 1: Chunk 1
Rank 2: Chunk 3	Rank 2: Chunk 12	Rank 2: Chunk 5
Rank 3: Chunk 5	Rank 3: Chunk 1	Rank 3: Chunk 3

We use: **Reciprocal Rank Fusion (RRF)**

$$RRF(d) = \sum_{j} \frac{w_j}{k + r_j(d)}$$

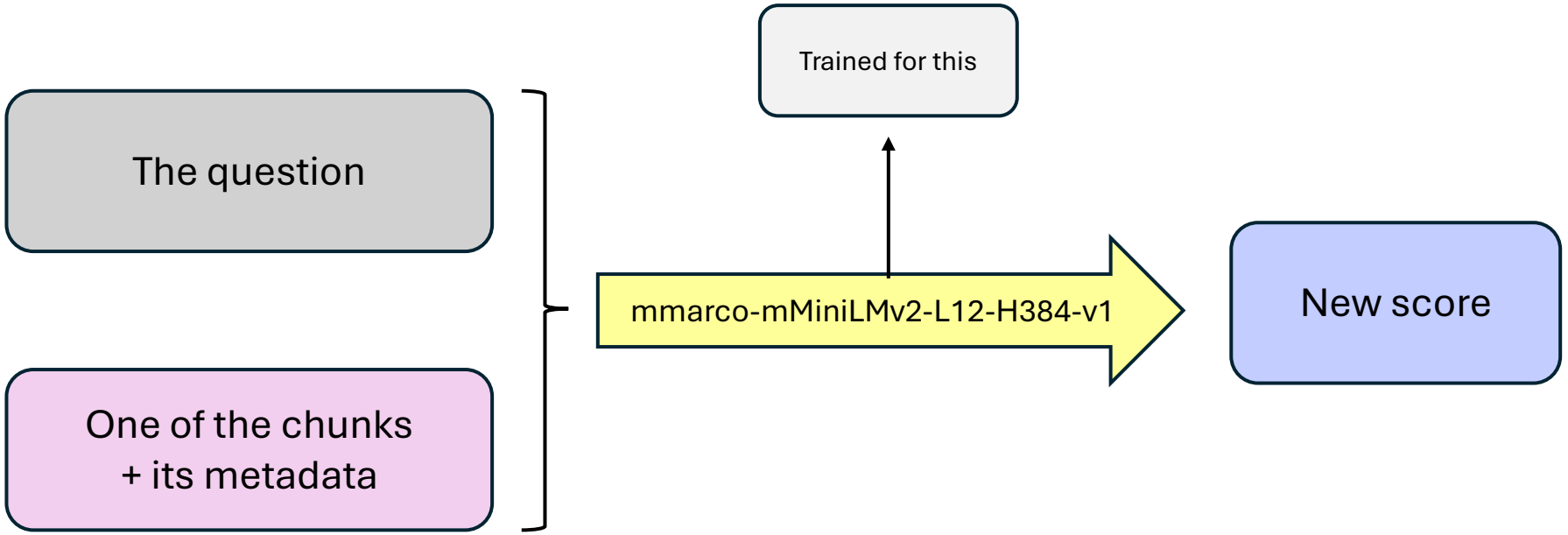
Choose the best 30 based on RRF

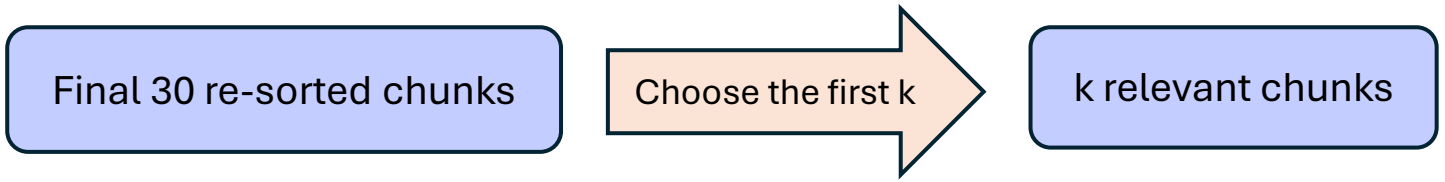
5- Retrieval

Choose the best 30 based on RRF

Making the chunk choice dépend ONLY on the RRF is simplistic. We use a smart RE-RANKER.

It is like a human who will check the chunks compared to the question and say if they are actually relevant or no.





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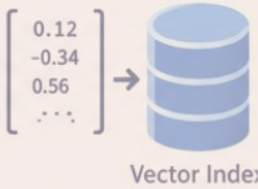
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Generate an answer grounded in the retrieved passages

Answer + page references

"The equipment is rated for 200°C … (pp. 12–14)."

RAG = Retrieve the right information first, then let the LLM **answer** using that evidence.

6- LLM Answer + Sources

The question

k relevant chunks

Only if given



Mistral Small 3.1

Retrieved chunk + Answer + Sources

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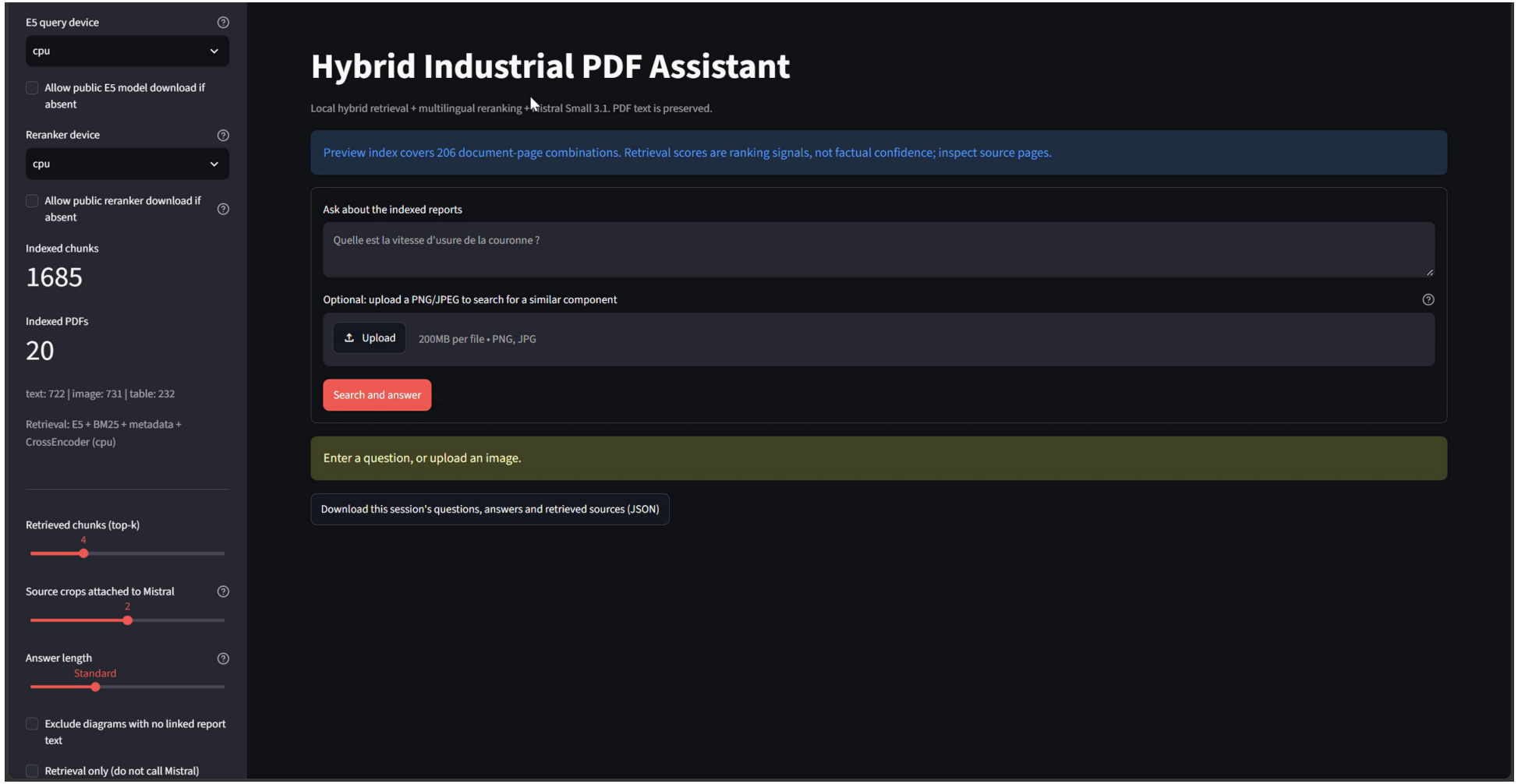
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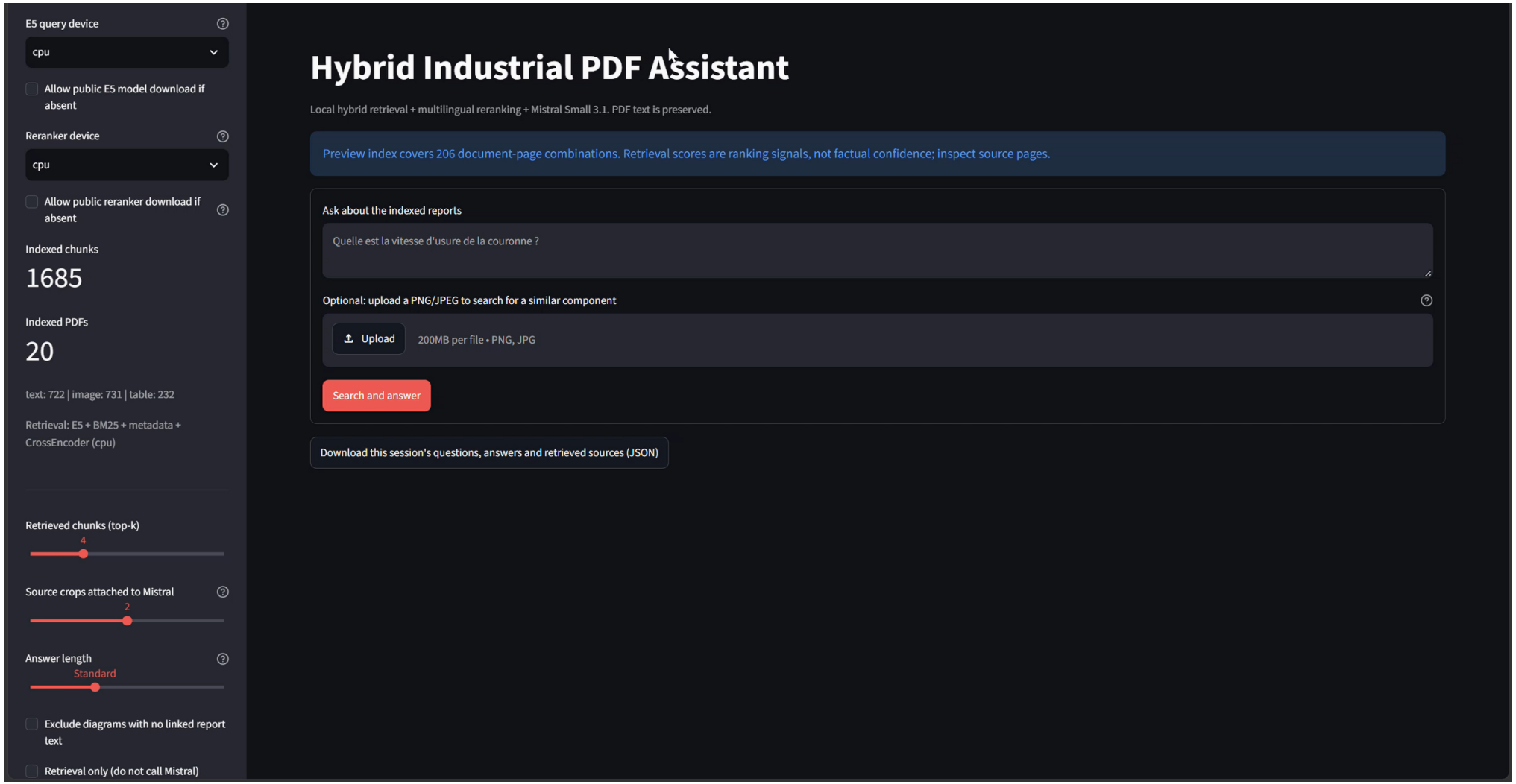
Example 1: Exact Numerical Extraction

Quelles sont les valeurs V1, V2 et V3 mesurées sur la couronne chez SIBELCO Compiègne ?



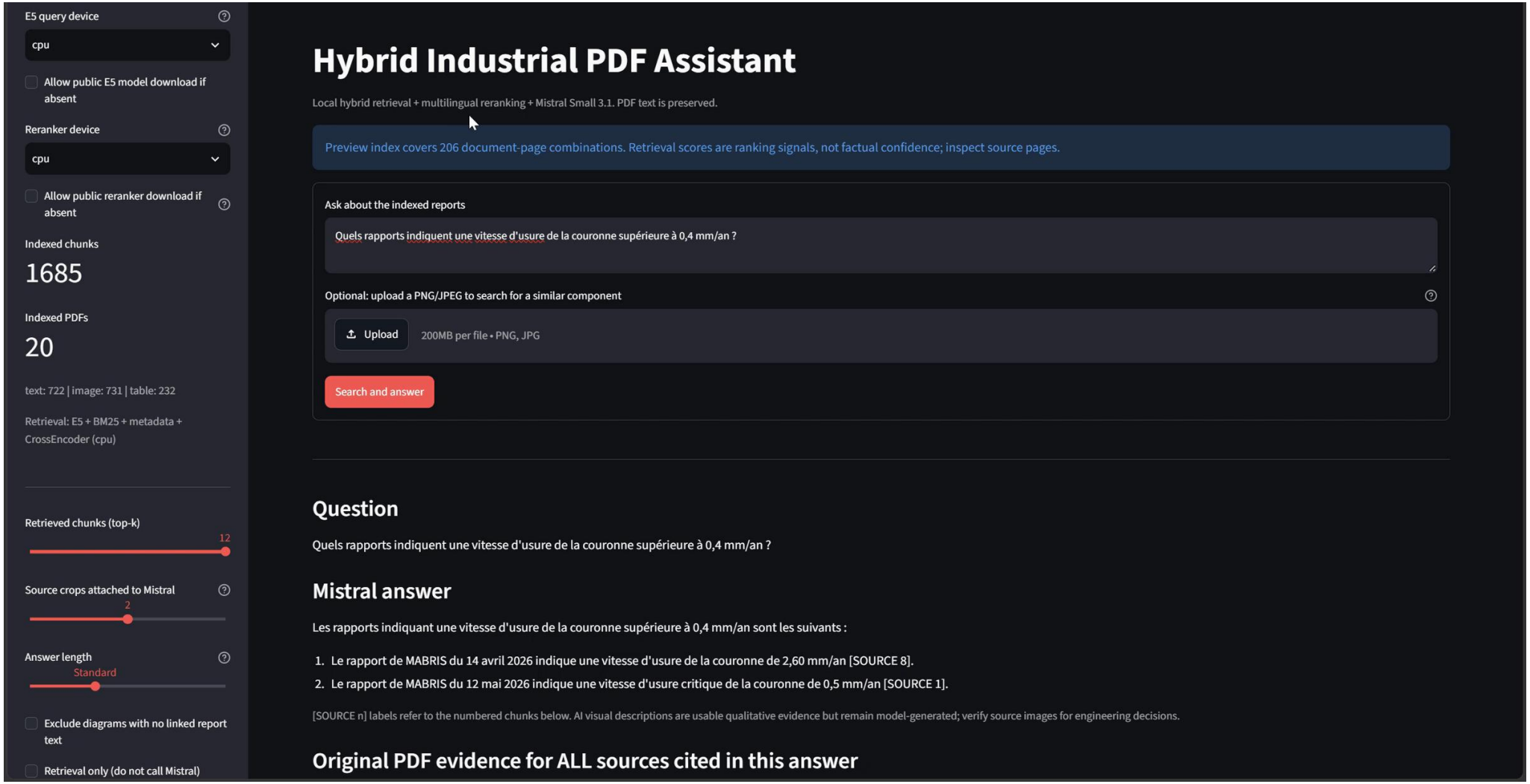
Example 2: Cross-report and multimodal search

Quels rapports indiquent une vitesse d'usure de la couronne supérieure à 0,4 mm/an ?



Example 3: Visual retrieval

Retrouve-moi une photographie montrant une dent d'engrenage avec des signes visibles d'usure ou de dommage. Dans quel rapport se trouve-t-elle et que voit-on sur l'image ?



Example 4: Multimodal Comparison with Visual input

À partir de cette photo, retrouve des cas similaires dans les rapports et trouve l'écart d'usure pour chaque cas. Compare mon écart d'usure de 0.2 mm avec les résultats précedent.

Image given



